

ENAE464 - 0202

Lab01: Pressure Measurements of Inviscid Flow over Cylinder

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Dr. Silbaugh, 02:00 PM

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In general, the letter should be addressed to the person or organization that requested the information contained in your report. For the purpose of this course, however, your letter can be addressed to the laboratory professor whose experiment you just have completed.

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1 Abstract

2 Nomenclature

3 Introduction

For the inviscid-flow model used in this lab, we assume a potential-flow field that is:

- **Incompressible (continuity):** $\nabla \cdot \vec{V} = 0$.
- **Irrotational:** $\nabla \times \vec{V} = 0$.

If $\nabla \times \vec{V} = 0$, then a *velocity potential* ϕ exists such that $\vec{V} = \nabla \phi$. Substituting into continuity gives Laplace's equation:

$$\nabla^2 \phi = 0 \quad \left(\text{i.e., } \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0 \right).$$

In order to model flow over a 2D cylinder, we use an approach presented in Anderson's Fundamentals of Aerodynamics, which uses a superposition of a uniform flow and a doublet flow, as presented in the figure below:

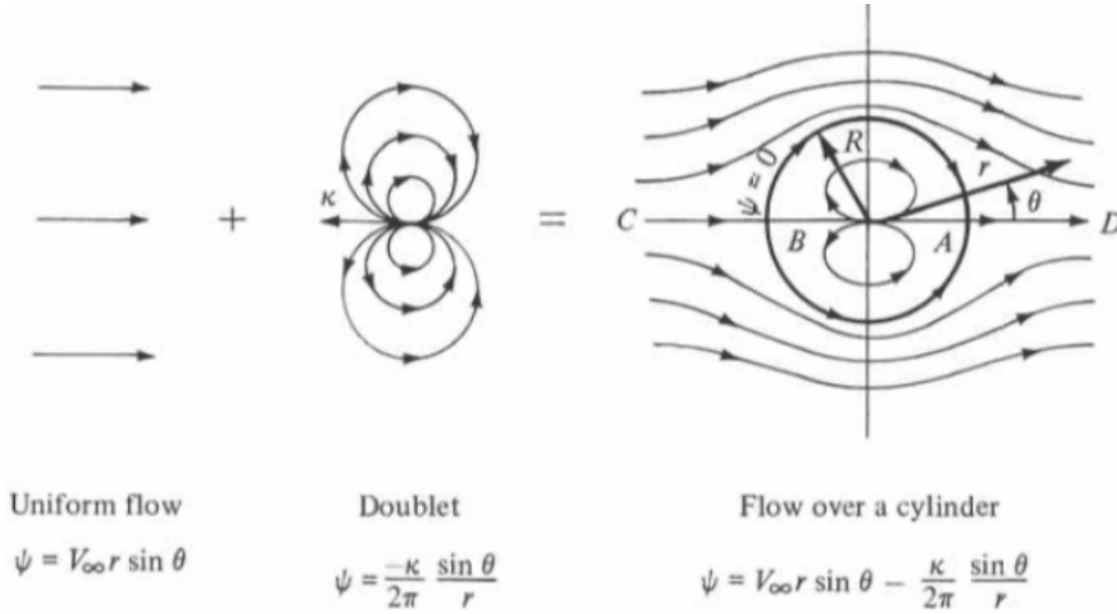


Figure 3.26 Superposition of a uniform flow and a doublet; nonlifting flow over a circular cylinder.

Figure 1: Superposition of uniform flow and a doublet to model nonlifting potential flow over a circular cylinder.

In polar coordinates (r, θ) , the stream function and velocity potential for both the uniform flow and the doublet are given by:

$$\phi_U = V_{\infty} r \cos(\theta) \quad (\text{velocity potential: uniform flow}) \quad (1)$$

$$\psi_U = V_{\infty} r \sin(\theta) \quad (\text{stream function: uniform flow}) \quad (2)$$

$$\phi_D = \frac{\kappa}{2\pi} \frac{\cos(\theta)}{r} \quad (\text{velocity potential: doublet}) \quad (3)$$

$$\psi_D = -\frac{\kappa}{2\pi} \frac{\sin(\theta)}{r} \quad (\text{stream function: doublet}) \quad (4)$$

where V_∞ is the freestream speed and κ is the doublet strength. Superposing the uniform flow and doublet stream functions gives

$$\begin{aligned} \psi &= \psi_U + \psi_D \\ &= V_\infty r \sin(\theta) - \frac{\kappa}{2\pi} \frac{\sin(\theta)}{r} \\ &= V_\infty r \sin(\theta) \left(1 - \frac{R^2}{r^2} \right), \end{aligned} \quad (5)$$

where the cylinder radius is defined by

$$R^2 \equiv \frac{\kappa}{2\pi V_\infty}.$$

The corresponding velocity field in polar coordinates follows from the stream function relations $V_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta}$ and $V_\theta = -\frac{\partial \psi}{\partial r}$:

$$V_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = \frac{1}{r} (V_\infty r \cos(\theta)) \left(1 - \frac{R^2}{r^2} \right) = \left(1 - \frac{R^2}{r^2} \right) V_\infty \cos(\theta), \quad (6)$$

$$V_\theta = -\frac{\partial \psi}{\partial r} = -\left(1 + \frac{R^2}{r^2} \right) V_\infty \sin(\theta). \quad (7)$$

At the surface of the cylinder, $r = R$, the velocity field is:

$$V_r = \left(1 - \frac{R^2}{R^2} \right) V_\infty \cos(\theta) = 0, \quad (8)$$

$$V_\theta = -\left(1 + \frac{R^2}{R^2} \right) V_\infty \sin(\theta) = -2V_\infty \sin(\theta). \quad (9)$$

This aligns with the assumption that the velocity normal to the stationary surface must be zero, and due to polar coordinates, the velocity normal to the surface is given by V_r . Since V_θ is perpendicular to the surface, due to construction of the coordinate system, it's equivalent to a tangential velocity, and since normal velocity is zero, $V = V_\theta = V_\infty \sin(\theta)$.

The pressure coefficient is given by:

$$C_p = 1 - \frac{V^2}{V_\infty^2}$$

$$V = -2V_\infty \sin(\theta) \quad (10)$$

$$\frac{V^2}{V_\infty^2} = 4 \sin^2 \theta \quad (11)$$

$$\boxed{C_p = 1 - \frac{V^2}{V_\infty^2} = 1 - 4 \sin^2 \theta} \quad (12)$$

4 Experimental Apparatus and Procedures

4.1 Experimental Apparatus

4.2 Operating Technique

5 Results

5.1 Operating Conditions

Table 1: Ambient Room Conditions during Experiment

Quantity	Value	Uncertainty
Room Temperature, T	25 °C	± 0.5 °C
Room Pressure, P	1009.0 hPa	–

The density of air (ρ) can be calculated using the Ideal Gas Law:

$$\rho = \frac{P}{RT}$$

where:

- P is the absolute pressure (in SI units: Pa)
- T is the absolute temperature (in Kelvin)
- R is the specific gas constant for dry air, $R = 287.05 \text{ J}/(\text{kg} \cdot \text{K})$

Given measurements:

$$P = 1009.0 \text{ hPa} = 100900 \text{ Pa}$$

$$T = 25 \text{ °C} = 25 + 273.15 = 298.15 \text{ K}$$

$$R = 287.05 \text{ J}/(\text{kg} \cdot \text{K})$$

$$\rho = \frac{100900 \text{ Pa}}{287.05 \text{ J}/(\text{kg} \cdot \text{K}) \times 298.15 \text{ K}}$$

Calculate:

$$\begin{aligned} \rho &= \frac{100900}{287.05 \times 298.15} \\ &= \frac{100900}{85542.5} \\ &\approx 1.18 \text{ kg/m}^3 \end{aligned}$$

Thus, the ambient air density during the experiment is:

$$\boxed{\rho = 1.18 \text{ kg/m}^3}$$

5.2 Measurements

5.3 Quantities of Interest

The theoretically predicted drag coefficient for a cylinder in inviscid flow is zero. This result arises because, under the assumption of inviscid (frictionless) flow, the skin friction coefficient c_f is zero, and the symmetry of both the cylinder and the flow ensures that the pressure distribution is identical on the front and back surfaces. As a result, the opposing pressure forces cancel each other, leading to no net drag. This phenomenon is known as **D'Alembert's Paradox**.

Performing numerical analysis using the equations from the lecture:

$$C_D = \frac{D}{\rho_\infty}$$

$$D = R \int_0^{2\pi} P \cos(\theta) d\theta$$

Due to cylinder symmetry:

$$D = R \int_0^{2\pi} P \cos(\theta) d\theta = 2R \int_0^\pi P \cos(\theta) d\theta$$

To numerically calculate the integral, we used the trapezoidal approximation, and the results of C_D vs. θ are shown in the figure below:

6 Analysis and Discussion

6.1 Data Conversion

6.2 Interpretation

As can be seen in the figure, the actual coefficient of drag is not zero. This is due to the fact that in real world air is viscous and its coefficient of friction $c_f \neq 0$. This combined with potential flow detachment leads to a non-zero drag coefficient.

7 Summary and Conclusions

8 References

References

- [1] J. D. Anderson and C. P. Cadou, *Fundamentals of Aerodynamics*, 7th ed. McGraw-Hill Education, March 17, 2023, pp. 209–324, ISBN: 9781266076442.

9 Appendices

9.1 Data

./data/pressure_vs_alpha.csv				
	α [°]	P_{∞} [cm]	P_0 [cm]	P [cm]
01	0	15.4	23.0	23.0
02	5	15.4	23.1	22.7
03	10	15.4	23.1	22.3
04	15	15.4	23.2	21.3
05	20	15.3	23.1	19.9
06	25	15.4	23.1	18.6
07	30	15.4	23.2	16.9
08	35	15.4	23.1	15.0
09	40	15.4	23.1	13.1
10	45	15.4	23.1	11.2
11	50	15.3	23.1	9.4
12	55	15.3	23.1	8.4
13	60	15.4	23.1	7.4
14	65	15.4	23.1	7.0
15	70	15.4	23.2	6.9
16	75	15.4	23.2	7.6
17	80	15.4	23.2	7.8
18	85	15.4	23.1	8.0
19	90	15.4	23.1	8.1
20	95	15.4	23.1	8.3
21	100	15.4	23.2	8.3
22	105	15.4	23.1	8.2
23	110	15.4	23.1	8.1
24	115	15.4	23.1	8.1
25	120	15.4	23.1	8.1
26	125	15.4	23.1	7.9
27	130	15.4	23.1	7.8
28	135	15.4	23.1	7.8
29	140	15.4	23.1	7.7
30	145	15.4	23.1	7.6
31	150	15.4	23.1	7.7
32	155	15.4	23.1	7.8
33	160	15.4	23.0	8.0
34	165	15.4	23.1	8.0
35	170	15.4	23.1	8.2
36	175	15.4	23.1	8.2
37	180	15.4	23.1	8.2

Table 2: Raw Measurements of Pressure Taps vs. Angle of Attack

9.2 Code